Cold Chain Shipping Procedure

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Purpose

This procedure defines how temperature-sensitive biologics (labelled storage 2-8 C) are packed, shipped, monitored and released. It applies to all lanes from the manufacturing site to distribution centres and hubs.

Packaging Configurations

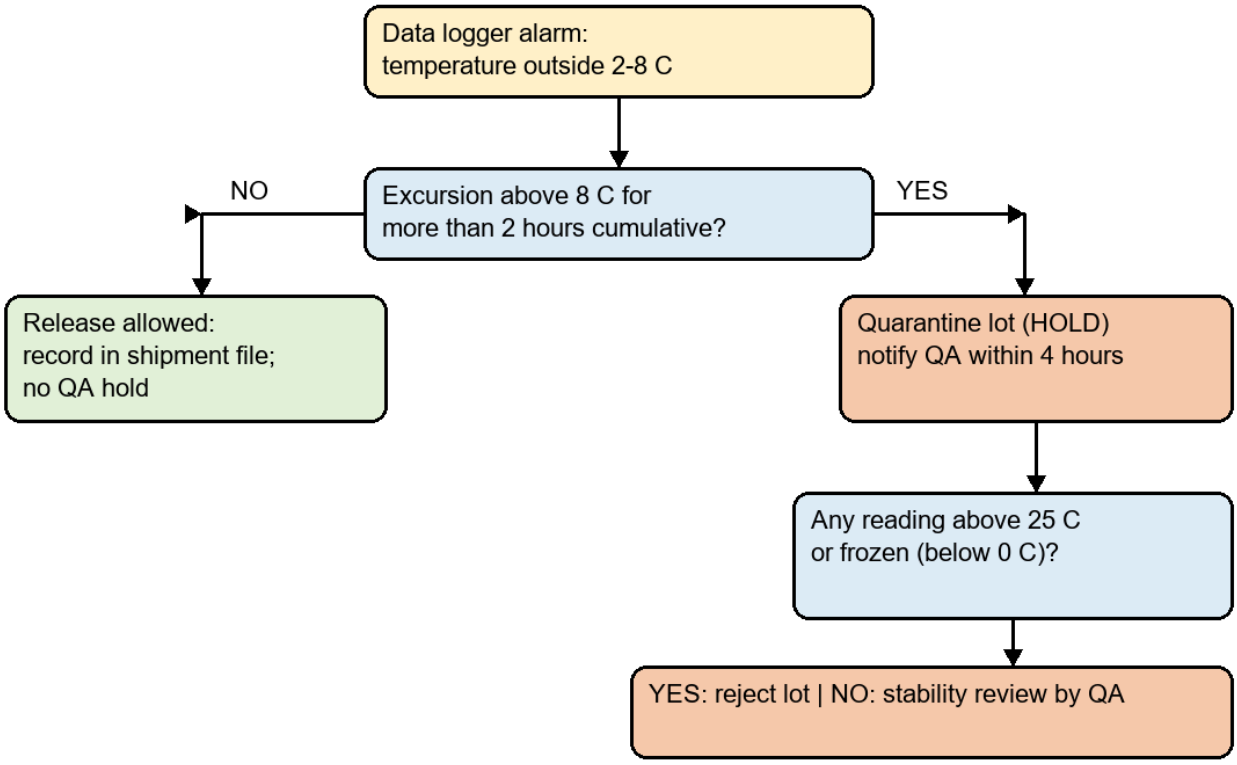
Select the packaging configuration from the table below. Summer configuration applies June to September.

Configuration	Qualified duration	Coolant	Use for
PCM shipper 96h	96 hours	4 PCM panels (5 C)	Lane L1, winter L2
PCM shipper 120h	120 hours	6 PCM panels (5 C)	Lane L2
Active container	unlimited (powered)	compressor unit	Lane L3, > 200 vials

Monitoring

Every shipper carries a calibrated data logger recording every 10 minutes. Loggers are downloaded at receipt and the file is attached to the shipment record before release.

Figure 1 - Temperature Excursion Decision Flow (2-8 C products)



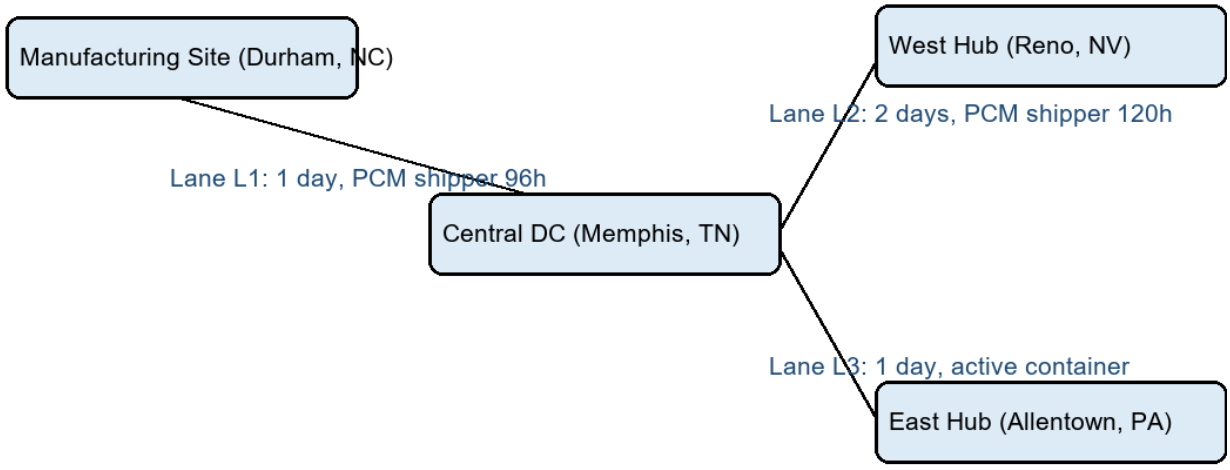
Excursion Handling

When a logger alarms, follow the decision flow in Figure 1. QA decides on release, quarantine or rejection; warehouse staff must not release a lot with an open excursion.

Distribution Network

The network and lane-specific packaging are shown in Figure 2.

Figure 2 - Cold Chain Distribution Network



All lanes: data logger every 10 minutes; summer lanes Jun-Sep use +20% coolant mass.

Records

Shipment records, logger files and excursion assessments are retained for 7 years.